

Seeding QDArchive

(5 / 10 ECTS Seminar and / or Project)



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QDArchive



QDArchive

A web service for researchers to publish and archive qualitative data

- With an emphasis on qualitative data analysis (QDA) files

Qualitative data = interview transcripts, research articles, audio/video, etc.

- QDA files = Structured data that capture the meaning in the qualitative data

Both qualitative data files and QDA files are useful for RAG creation

- And LLM training

Seeding QDArchive

QDArchive is new and under active development (still a prototype)

Chicken and egg problem = how to attract researchers?

Answer = seed QDArchive with available open qualitative data

This project is to lay the groundwork for seeding QDArchive

Project Course



Seminar / Project Timeline

Part 1: **Data acquisition** (5 ECTS)

- ~~Until March 15th~~ Until April 17th

Part 2: **Data classification** (5 ECTS)

- ~~Until April 12th~~ Until July 17th

Ways of Participating and Acquiring ECTS

- 5 ECTS Seminar (generic name “Applied Software Engineering Seminar”)
 - Acquisition only
 - Classification only
 - Analysis only
- 10 ECTS Project (generic name “Applied Software Engineering Project”)
 - Acquisition + classification
 - Classification + analysis
 - ~~Acquisition + analysis~~

Coordination Meetings

Meet weekly Fri 4pm CET in our [public lecture hall](#)

~~Acquisition start = Feb 13 (Fri), 4pm CET~~

- ~~• Project meetings on Feb 20 (Fri), Feb 26 (Thu), Mar 6 (Fri), Mar 13 (Fri)~~
- ~~• Always at 4pm CET in our [public lecture hall](#)~~

~~Classification start = Mon Mar 16, 4pm CET~~

- ~~• Three weekly meetings Mondays Mar 23, 30, Apr 6 at 4pm~~

~~Analysis start = Fri Apr 17, 5pm CET~~

- ~~• Relaxed, until end of teaching time, schedule TBD then~~

Part 1: Acquisition



Part 1 Step 1

Find as much qualitative **research** projects on the web as you can

- Start with finding qualitative data research projects
 - Can be identified by containing Analysis Data files (QDA files)
 - Or may simply have a description that says qualitative (data) research
- Try to find **Analysis Data files** (not just qualitative data)
 - Everything can be interpreted as qualitative data
 - We don't want you (to try) downloading arxiv
- Collect as much metadata as possible with the qualitative data
 - **The data must have an open license**
 - So ensure you get the license
- **Basic heuristic: Just download the research project folder (all files)**

A Qualitative (Data) Research Project is a Web Folder

QDA Research Project Data Set (website or folder or archive)

Analysis Data file (a.k.a. QDA file)

- The key data structure (file) we are interested in; result of a researcher analyzing a domain
- Possible extensions: qdpx, mx24, etc. see file extensions spreadsheet
- Number of files: 0 or 1 (most likely), but also possible that there is more than 1 file
- Analysis files may contain none, some, or all primary data files

Primary Data files

- The input data to a researcher's analysis, for example, interview transcripts or research papers
- Possible extensions: pdf, doc, docx, txt, jpg, etc. (anything goes)
- Number of files: 0..n (if 0, hopefully the primary data files are included in the analysis data file)
- Primary data files may be included in an analysis data file and may also be standalone files in the research project

Additional Data files

- Other additional files the researcher considered important (that are not analysis or primary data files)
- Possible extensions: pdf, doc, docx, txt, jpg, etc. (anything goes)
- Number of files: 0..n (no expectations)
- Please also collect them

Part 1 Analysis Data File (QDA File) Types

An analysis data file is structured data that

- A scientist creates when interpreting primary data like interview transcripts

We have a particular interest in qualitative data analysis (QDA) files, e.g.

- qdpx (called the REFI format, see <https://www.qdasoftware.org/>)

[Spreadsheet with known file types / extensions](#)

- Please add to it if you find new information

Part 1 Primary Data File Types

A primary data file is a file that contains any form of qualitative data like

- Interview transcripts or research articles

File types include

- txt, PDF, RTF, DOCX, ...

Part 1 Additional Data Files

In the scope of a qualitative research project there can be other files that are

- Not analysis or primary data files
- But something the researcher thinks belongs to the project

Part 1 Qualitative Data Repositories

Qualitative research projects are often archived in repositories

- [Data sources to start with](#)

Part 1 Open Data Licenses

- Any Creative Commons license
- Many others...

It will be more difficult to find a license

- No license = close proprietary do not use

Part 1 Step 2

Build a pipeline to download all this data

- Both files as well as metadata
 - Put metadata into an associated database (exportable to a CSV file)
- The database schema will be evolved during the project
 - Source, source link, license information, project scope, ...
- If you can't fill a particular field, leave it empty

Part 1 Step 3

Operate the pipeline to create your own archive

Part 1 Ongoing



Report about technical challenges with data (not with programming)

Part 1 Submission / Delivery

Work is due by April 17th ~~March 15th~~

Tag your git version as part-1-release

Fill in a form, which asks about

- Download link for database
- Link to folder with downloaded files

Part 2: Classification



Part 2 Step 1

~~Merge all student databases into one (for yourself) to work on~~

- ~~• Identify and remove duplicates, if possible~~

Optionally, you can merge your data with other students

Filter all projects into one of

- QDA_PROJECT
- QD_PROJECT
- OTHER_PROJECT
- NOT_A_PROJECT

Add this information to a new column “type” of type PROJECT_TYPE

Part 2 Step 1 Continued

Use the file types to derive the PROJECT_TYPE

- QDA_PROJECT if there is a file with QDA file extension
- QD_PROJECT if not a QDA_PROJECT and there are primary data files
- OTHER_PROJECT if not a QD_PROJECT and there are valid data files
- NOT_A_PROJECT if nothing can be derived about file types

Part 2 Step 2

Develop a classifier for the available data

- Uses both the base data (the files) and the metadata

Use the [ISIC Rev. 5 standard](#) [1] for a hierarchical classification taxonomy

- Go down two levels, i.e. divisions (not just sections)

Also consider creating tags for searching

Part 2 Step 3

Run the classifier on your data by project type

- Once for QDA_PROJECT types of project
 - Classify the project (as the sum of its files) itself
 - Classify each primary data file as well
- Once for QD_PROJECT types of project
 - Classify the project (as the sum of its files) itself
 - Classify each primary data file as well

And by repository

Part 2 Results Step 4a

Commit your sqlite database to your repository under name

- student_id-sq26-classification.db

Tag the database with label classification-results

Part 2 Results Step 4b

Report about the resulting statistics, by repository

- Which project types were found, how many
- What is the dominant class

Report by filling in this form (once for each repo you worked on)

- <https://forms.gle/wxTGQFBQbBvFi3N69>

Part 2 Results Step 4c

Extract a simple table from your database, consisting of the following columns

- repository_id
- project_type
- project_title
- primary_class
- secondary_class // if any
- no_project_files // No files in project in total

Submit the table as an XLSX spreadsheet on moo.uni1.de

- See the results section

Cut-off date is as announced

Total Number of Distributions to Report About

		Project type	
		QDA_Project	QD_Project
Repository	1	Distribution 1	Distribution 2
	2	Distribution 3	Distribution 4
	...	...	...

Part 2 Results Step 4d

Create a PDF (the report) summarizing the results, with the following structure

1. Repository

- a. Histogram of primary classes identified
 - i. Use the full class name as the bin name (find them in the shared folder) [1]
 - ii. Add the count as a number on top of the visualizing bar
 - iii. Please use vector graphics for any graphics so that one can zoom in
- b. Rank-ordered list of classes in table format, starting with most common class
 - i. Include only the top twenty classes
 - ii. For each class, provide the count
- c. Any comments you may have on your findings

2. Continue with next repository

Submit PDF on moo.uni1.de

[1] https://docs.google.com/spreadsheets/d/1rHo_1WJJENXFKkLW8-GWmzq-fxmNzD_E/edit?usp=drive_link&ouid=117127688950812758222&rtpof=true&sd=true

Part 2 Ongoing

Report about technical challenges with data (not with programming)

Thank you! Any questions?



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